Validation-Budget Sensitivity in Lightweight Link Prediction: An Empirical Study

PILOT MANUSCRIPT — NOT YET A SUBMISSION-READY SCI FINAL VERSION

Abstract

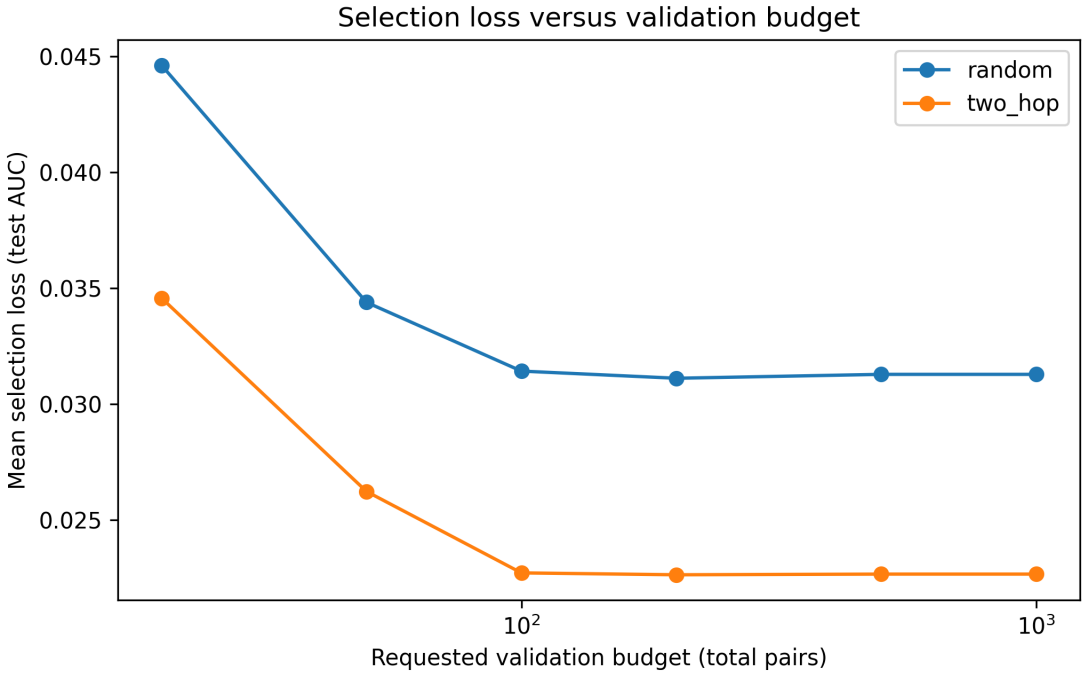
This pilot measures how a limited validation budget affects downstream algorithm selection among six lightweight link-prediction heuristics. Methods are selected on budgeted validation samples and evaluated on independent test sets. Selection loss is the gap between the selected method and the test-optimal method. The bundled pilot generated 1,584 selection experiments across twelve graphs and two negative-sampling regimes. Results show lower pooled mean selection loss at larger requested budgets, while high-budget points are partly constrained by small validation pools. The package is a reproducible protocol and pilot, not a final submission-ready empirical study.

Pilot Results

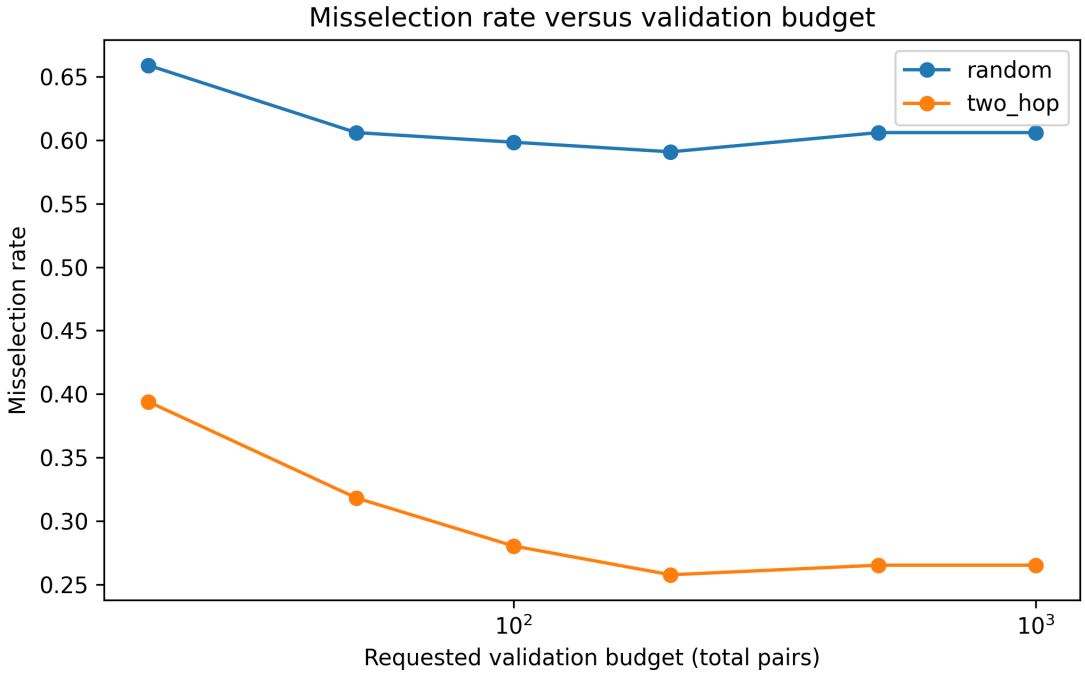
Negative regime	Budget	Mean loss	Misselection
random	20	0.0446	0.659
random	50	0.0344	0.606
random	100	0.0314	0.598
random	200	0.0311	0.591
random	500	0.0313	0.606
random	1000	0.0313	0.606
two_hop	20	0.0346	0.394
two_hop	50	0.0262	0.318
two_hop	100	0.0227	0.280
two_hop	200	0.0226	0.258
two_hop	500	0.0227	0.265
two_hop	1000	0.0227	0.265

Interpretation: the pilot shows that validation budget can alter method selection and independent-test regret. The observed high-budget plateau should not be treated as a universal law because some graphs cannot provide the requested number of balanced validation pairs.

Selection loss versus validation budget



Misselection rate versus validation budget



Submission Readiness

- Replace or augment the pilot with a larger documented benchmark suite.
- Freeze the final protocol before the full run.
- Add network-level uncertainty intervals and sensitivity checks.
- Update the manuscript strictly from final experiment outputs.
- Recheck current journal scope, indexing, quartile and publication policy before submission.